

HoshyarInvestAI

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Chapter 1

Introduction

HoshyarInvestAI is an AI-powered investment platform designed to help investors in Pakistan's tech sector by providing real-time market insights, sentiment analysis, and risk management tools. It leverages NLP and advanced data analysis to deliver actionable investment suggestions tailored to user preferences. [1].

1.1 Problem Statement

Investing in the rapidly growing tech sector in Pakistan comes with great implications. Traditional models still depend on the likes of revenue and market size without considering, and therefore losing on the important aspects of the tech industry such as innovation, scale and positioning which are core to any technology driven industry. Further, investors in tech in Pakistan do not possess any real-time AI solutions that bridge the market, financial data and sentiment with actionables. This missing market means losing at cornering opportunities and analysis of various sectors especially the technology sector where change happens rapidly. Our project solves this issue as it involves the creation of an investment bot that is powered by AI and factors in the trends within the top publicly listed tech companies in Pakistan. Using advanced forms of NLP and semantic analysis, the system will give real time investment suggestions, risk management profiles and trend forecasts allowing for active investor decision making in an ever changing investment environment.

1.2 Scope

The ultimate intent of HoshyarInvestAI is to collect and integrate current data-it can be financial documents or trends, public opinion, or statistics, so specifically it presents rel-

evant trends related to the tech industry within Pakistan-meaning to keep users abreast of all changes recently made. Such a type of information is processed in the system with the aid of deep learning algorithms and Natural Language Processing (NLP) techniques for trend analysis and stock predictions as well as investment risk appraisal. Such adjustments are made to suit the nature of the technology market in Pakistan. Individual users get buy,hold, and sell investment signals regarding the level of risk they can tolerate and their choice. The platform does comprise an analytics module that allows the users to analyze the portfolios and the situation in the market in real time. Also, the platform includes tools for back trading investing tools on a simulated environment within the functionalities of historical data analysis. The aim of such a platform is to enable informed and intelligent decision making focusing on investments in the technology industry in Pakistan.

1.3 Modules

Write down the modules of the proposed project. Each module should highlight features, using bulleted/numbered notation. When developing both a mobile app and a web app, group the modules according to the system types, such as, Client Web App, Client Mobile App, Admin Web App etc.

1.3.1 Module 1: Data Collection and Integration

More specifically, this module targets the automation of collection and integration of live financial data with country and world market trends, expectations of the actors in the market, and important economic indicators. plenty of information from stakeholders like the Pakistan Stock Exchange (PSX), published financials and the news etc. will be collected in order to achieve full coverage.

1.3.1.1 Features:

1. Data acquiring in real-time through PSX and financial reports.
2. Data populating through news/journal sources and market sentiment analysis instruments.

1.3.2 Module 2: Feature Engineering

This module transforms the raw data into features so we can use it for model training. Its main features are given below,

1.3.2.1 Features:

1. Data Preprocessing
2. Feature Extraction
3. Feature Selection
4. Encoding and Scaling
5. Feature Engineering for Personalization
6. Feature Validation and Testing

1.3.3 Module 3: Advanced Analysis and Modelling

This module terms the input data which has been accumulated through this internet interface and advances the application of deep learning models and high level feature engineering. The emphasis is on processing and cleaning the raw information, creating features and building models of stock prices and movements in the markets.

1.3.3.1 Features:

1. Detailing features to find critical financial features and their trends.
2. Utilization of statistical model EGARCH for risk assessment.
3. Use of reinforcement learning model for user behavior.
4. Use of Temporal fusion transformer model for stock prediction.

1.3.4 Module 4: Dashboard and User Interface

This module offers a graphical user interface (GUI) which exactly addresses the investor's needs. It contains a depiction of the data, the performance of the system over time, along with a simple interface for handling investments as well as market conditions.

1.3.4.1 Features:

1. Visualization of dynamic information and monitoring of one's investment portfolio.

2. Options of multiple tech companies to invest in, along with visualization of their stocks.

1.3.5 Module 5: Backtesting and Strategy Validation

This module offers the possibility to the users to review how their investment strategies would have worked historically. It's a module that explains how different strategies have been used over time and their efficacy.

1.3.5.1 Features:

1. Strategy validation through historical data simulation.
2. Investment guide's strategy performance review and suggestions for improvement.

1.4 User Classes and Characteristics

User classes are stated below

User class		Description
Investor		Tech investors in Pakistan who will use the platform to make data-driven decisions. They will vary in experience, from novices to experts.
Admin		Administrators managing system performance, data updates, and user access. They require technical knowledge to ensure smooth operations.
Data Analysts	Ana-	Analysts who will use the system to interpret data and refine investment strategies. They will interact with the platform for in-depth analysis.

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

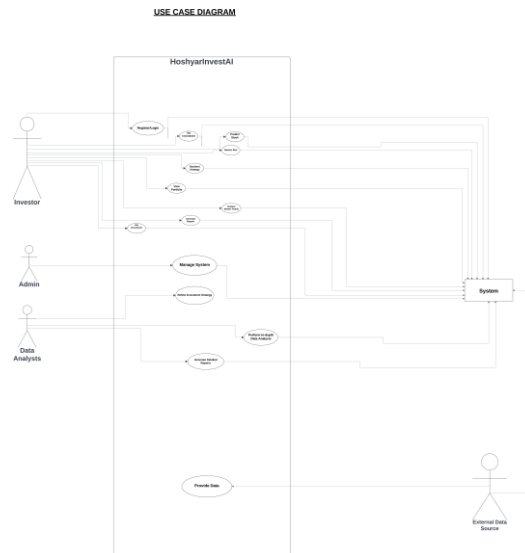


Figure 2.1: UseCaseDiagram

2.2 Detailed UseCase

1- View Real-Time Stock Data

Use Case	View Real-Time Stock Data
Designation	UC-001
Author	Saim
Priority	High
Actor	Investor
Type	Primary
Description	Investors can view live stock data of tech companies from multiple sources.
Pre-Condition	• The investor has an active account in the HoshyarInvestAI system. • The investor is logged into their HoshyarInvestAI account.
Post-Condition	The investor successfully views up-to-date stock data and financial details.
Normal Flow	• The investor logs into their HoshyarInvestAI account. • The system presents the investor's dashboard. • The investor selects the "Stock Data" option. • The system fetches the latest stock data from multiple tech companies. • The system displays live stock data, including price, volume, and trends. • The investor can filter or sort the stock data by company, price, or date. • The system updates the displayed data in real-time as the stock values fluctuate.
Alternative Flow	If stock data from a particular company is unavailable or an error occurs during data fetching, the system will display an appropriate error message and suggest trying again later or switching data sources.
Exception Scenarios	None

2-Receive Investment Recommendations

Use Case	Receive Investment Recommendations
Designation	UC-002
Author	Ayesha
Priority	High
Actor	Investor
Type	Primary
Description	The system provides personalized investment recommendations based on user preferences and market trends.
Pre-Condition	• The investor has completed their investment profile. • Market data is available.
Post-Condition	The investor receives accurate and timely recommendations.
Normal Flow	• The investor logs into HoshyarInvestAI. • The system analyzes user preferences and market trends. • The investor selects "Recommendations." • The system generates personalized recommendations. • The system displays recommendations for tech stocks. • The investor views detailed insights and can save or act on recommendations.
Alternative Flow	If data for recommendations is missing, the system suggests alternative companies or industries.
Exception Scenarios	None

3- Perform Backtesting on Investment Strategy

Use Case	Perform Backtesting on Investment Strategy
Designation	UC-003
Author	Murtaza
Priority	Medium
Actor	Investor
Type	Primary
Description	Investors can test their investment strategies on historical data to see how they would have performed.
Pre-Condition	• The investor has selected a strategy. • Historical data is available.
Post-Condition	The investor receives results on the backtested strategy's performance.
Normal Flow	• The investor logs into HoshyarInvestAI. • The system presents the backtesting feature. • The investor selects a strategy to test. • The system runs the backtest on historical data. • The system displays performance metrics (e.g., returns, risk). • The investor reviews the results and adjusts the strategy if needed.
Alternative Flow	If data is missing or incomplete, the system will use default datasets.
Exception Scenarios	None

4-Explore Interactive Visualizations

Use Case	Explore Interactive Visualizations
Designation	UC-004
Author	Saim
Priority	Medium
Actor	Investor
Type	Primary
Description	Investors can explore data insights through interactive charts, graphs, and heatmaps.
Pre-Condition	The investor has selected a dataset or strategy to visualize.
Post-Condition	The investor can interact with and manipulate the visual data.
Normal Flow	- The investor logs into HoshyarInvestAI. - The system displays the dashboard with visualization options. - The investor selects "Data Visualization." - The system presents interactive charts/graphs based on their portfolio or selected data. - The investor can interact (zoom, filter, etc.) with the visualizations. - The system updates the visualizations based on user input in real-time.fluctuate.
Alternative Flow	If no data is available for visualization, the system will display a notification and suggest another dataset.
Exception Scenarios	None

5- Manage User Profile

Use Case	Manage User Profile
Designation	UC-005
Author	Ayesha
Priority	Medium
Actor	Investor
Type	Primary
Description	Users can manage and update their profiles, including personal information and investment preferences.
Pre-Condition	• The investor is logged into their account.
Post-Condition	The investor's profile is successfully updated.
Normal Flow	• The investor logs into HoshyarInvestAI. • The system presents the investor's dashboard. • The investor selects "Profile Management." • The system displays the user's profile details. • The investor updates their information (e.g., name, contact, preferences). • The system validates and saves the updated information. • The investor receives confirmation of a successful update.
Alternative Flow	If validation fails (e.g., invalid data), the system requests correction.
Exception Scenarios	None

6- Analyze Market Trends

Use Case	Analyze Market Trends
Designation	UC-006
Author	Murtaza
Priority	High
Actor	Investor, Data Analyst
Type	Primary
Description	The system analyzes real-time market trends for tech companies, providing insights for investors.
Pre-Condition	Market data is available.
Post-Condition	Market trends are displayed to the user in real-time.
Normal Flow	- The investor logs into HoshyarInvestAI. - The system fetches real-time market data. - The system analyzes the data using pre-trained models. - The investor selects "Market Trends." - The system displays trend analysis (e.g., growth, risk factors). - The investor views trends and insights and can act on the data.
Alternative Flow	If no market data is available, the system notifies the investor.
Exception Scenarios	None

7-Generate Risk Assessment Reports

Use Case	Generate Risk Assessment Reports
Designation	UC-007
Author	Murtaza
Priority	Medium
Actor	Investor, Data Analyst
Type	Primary
Description	The system generates detailed risk assessments for tech investments based on historical and real-time data.
Pre-Condition	• The investor has selected a company or portfolio to assess.
Post-Condition	The risk report is generated and displayed.
Normal Flow	• The investor logs into HoshyarInvestAI. • The investor selects "Risk Assessment." • The system analyzes the selected company or portfolio. • The system generates a risk assessment report. • The investor views detailed risk metrics (e.g., volatility, market sensitivity). • The investor can use the report to inform investment decisions.
Alternative Flow	If insufficient data is available for analysis, the system will provide a partial report or suggest alternative assessments.
Exception Scenarios	None

2.3 Functional Requirements

2.3.1 Module 1: Data Collection and Integration

This module handles the real-time data acquisition from multiple sources like PSX, financial reports, and sentiment analysis from news and journals.

1. The system shall gather real-time stock market data from Pakistan Stock Exchange (PSX).
2. The system shall gather financial reports from publicly listed companies.
3. The system shall analyze market sentiment using news and journal sources.
4. The system shall integrate world market trends and economic indicators.

2.3.2 Module 2: Feature Engineering

This module transforms raw data into features that are used for model training and predictions.

1. The system shall preprocess raw stock market and financial data.
2. The system shall extract relevant features for machine learning models.
3. The system shall allow feature selection based on investment strategies.
4. The system shall encode and scale the data for model optimization.
5. The system shall validate features through backtesting and simulation.

2.3.3 Module 3: Advanced Analysis and Modeling

This module applies machine learning and deep learning models to predict stock prices and assess risks.

1. The system shall utilize deep learning models to predict stock price movements.
2. The system shall generate a detailed risk assessment report for each stock.
3. The system shall identify critical financial trends for tech companies.
4. The system shall provide recommendations based on the risk level.

2.3.4 Module 4: Dashboard and User Interface

This module provides a user-friendly interface with real-time data visualization and monitoring tools.

1. The system shall visualize stock performance in a dynamic dashboard.
2. The system shall allow users to monitor their investment portfolio performance.
3. The system shall provide comparison options between different tech companies.
4. The system shall allow real-time interaction with market data through a graphical interface.

2.3.5 Module 5: Backtesting and Strategy Validation

This module allows users to validate their investment strategies by simulating historical data.

1. The system shall backtest historical stock data to validate investment strategies.
2. The system shall provide users with suggestions for strategy improvement.
3. The system shall simulate different market scenarios to assess strategy effectiveness.

2.4 Non-Functional Requirements

2.4.1 Reliability

1. The system shall have a mean time between failures (MTBF) of at least 1000 hours.
2. The system shall have failover mechanisms to recover data within 1 hour of failure.

2.4.2 Usability

1. The system shall allow users to view stock data with a single click.
2. The user interface shall require less than 10 minutes of training for new users.
3. The system shall have built-in error recovery mechanisms for incorrect inputs.

2.4.3 Performance

1. The system shall respond to user queries (e.g., stock trend analysis) within 3 seconds.
2. The system shall support up to 500 simultaneous user queries without performance degradation.

2.4.4 Security

1. The system shall encrypt all financial data in transit and at rest using AES-256 encryption.
2. The system shall require multi-factor authentication for all user logins.
3. The system shall log all user activities and notify the admin of suspicious behavior.

2.5 Domain Model

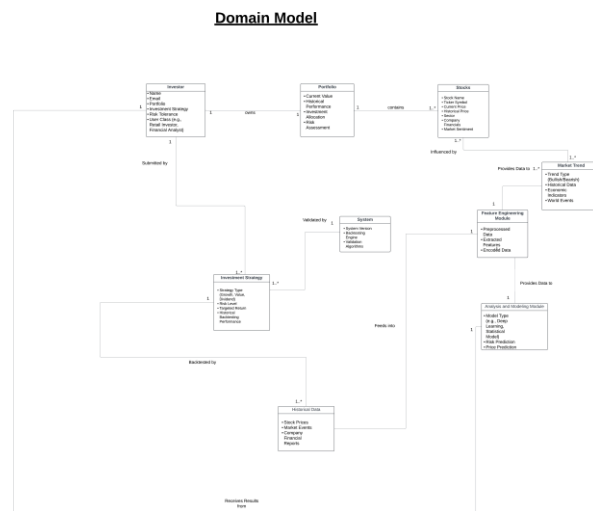


Figure 2.2: Domain Model

Chapter 3

System Overview

General Description

HoshyarInvestAI is an AI-driven investment bot with a prospect to invest in Pakistan's tech sector. It offers timely stock predictions, investment recommendations, and portfolio management tools to serve the needs of investors. It makes use of the data feeding in from multiple sources, such as financial reports and stock market data, and news articles with their respective sentiment analysis to provide investors with data-driven insights. The ultimate goal of the system is to assist individual and institutional investors in making effective choices regarding optimal investment strategies by using sophisticated machine learning models such as TFT and DRL.

Features: The system enables access to real-time data for stocks, one can analyze market trends based on past trends, robust backtesting of various investment strategies, and personalized recommendations tailored to specific market conditions and user needs.

Context: This is a system catering to fast-tech companies, which provides investment tools to optimize portfolios with insights beyond traditional investment metrics. Users may be on either the web application or mobile application, while administrators will be handling the backend data and models of the system.

3.1 Architectural Design

3.2 Design Models

Design Models for Object Oriented Development Approach

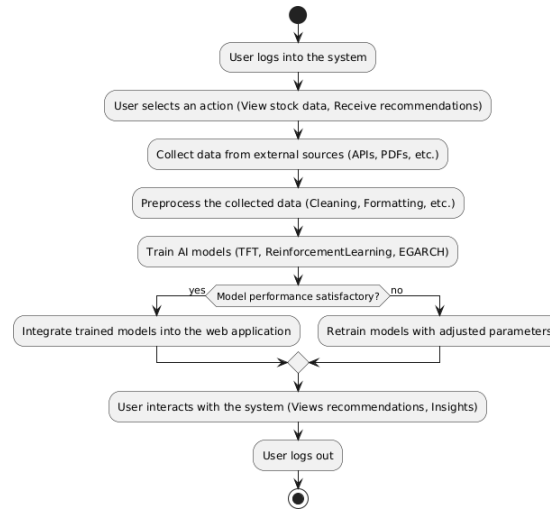


Figure 3.1: Activity Diagram

3.2.1 Activity Diagram

Description Activity Diagram for HoshyarInvestAI Major Activities:

User Start: A user begins his interaction with the system. User Login: The user logs into his account of HoshyarInvestAI Choose Action: At this step, he specifies his action to be executed-for example, to see data on a stock or get recommendations. Data Gathering: The system acquires data from external sources, such as APIs. Data Preprocessing: The system preprocesses the gathered data, cleaning and formatting it Train Models: The system trains models on the data preprocessed Evaluate Models: The system evaluates model performance. Decision: Is the model good enough? Yes: Deploy the model in the application No: Back to training of the model. User Interaction: User interacts with the system (looks at recommendations, visualizations). End: User logs out or closes the session.

3.2.2 Class Diagram

Description Investor: The class is the user and has methods to set up an investment strategy and to view the portfolio.

Associations: An investor has one portfolio An investor submits one or more investment strategies. Portfolio: It is a class that contains the current value, historical performance, as well as allocation details of the investor's portfolio.

Associations: A portfolio has multiple stocks Stock: It represents individual stocks. Each stock is subjected to the influence of a market trend.

MarketTrend: It depicts general market trend or trends (in both bullish and bearish modes)

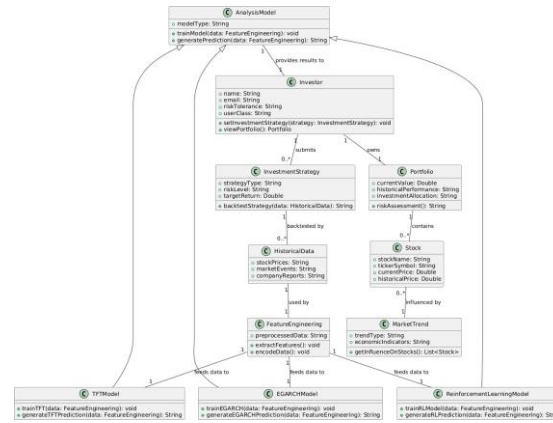


Figure 3.2: Class Diagram

that are acting upon multiple stocks.

InvestmentStrategy: These are the strategies that the investor follows with some strategy to check over the strategy on historical data.

Associations: The investment strategies are checked through historical data. **FeatureEngineering:** This module deals with data preprocessing and extraction of features which are needed for training the model. **AnalysisModel:** It denotes those AI models that can be used to predict stocks or those which can be trained to generate suggestions.

HistoricalData: This data includes historical prices and events, from stock prices and market events to financial reports, all are fed into the system for backtesting investment strategies and data feeding into feature engineering processes.

3.2.3 Class Level Sequence Diagram

Description

Investor login: The Investor logs in and the system authenticates the user **Request Investment Recommendations:** The Investor requests investment recommendations **Preprocess Data:** The system invokes the FeatureEngineering class to preprocess data. **Fetches HistoricalData** is passed to FeatureEngineering for preprocessing. **Model Training and Predictions:** **TFT:** If the TFT model is selected, the system trains the model and retrieves the prediction. **Reinforcement Learning:** In case the RL model is chosen the system trains the RL model and fetches the prediction. **EGARCH:** In case the EGARCH model is chosen the system trains the model and fetches the prediction. **Show Results:** The system finally shows investment recommendations to the Investor. **Key Features:** Alt blocks represent all the different branches of the sequence in which different models may be used (TFT, Reinforcement Learning or EGARCH). The investor interacts with the system to make requests for prediction, which in turn train the appropriate models.

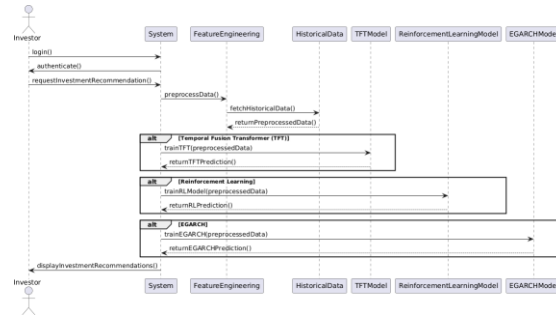


Figure 3.3: ClassLevelSequenceDiagram

3.2.4 State Transition Diagram

Description

Here's an example of tracking the Investor state transition while soliciting investment advice and processing the same in the back-end data gathering, model training, and the generation of output within the system.

Main States: Logged Out: This is the initiation state whereby the user is not logged into the system.

Event: User logs in → Transition to "Logged In" state. **Logged In:** the states can log in and interact with the system.

Event: User creates forecasts for shares → Enters the "Processing Request" state. **Processing Request:** The system collects data and processes the data while training the models.

Event: Data and models are successfully processed with no errors → Transitioned to "Recommendation Ready" state. **Event:** The processing of data fails → Transitioned to "Error Handling" state. **Recommendation Ready:** The system has successfully generated an investment recommendation through the models.

Event: The user has accessed their recommendations → Transitioned to "Displaying Recommendations" state. **Error Handling:** The error has been incurred in the context of data retrieval or model training.

Event: Retry or correct error → Return back to "Processing Request" state. **Rendering Recommendations:** The recommendations are rendered to the user.

Event: User logs out → Transition to "Logged Out" state.

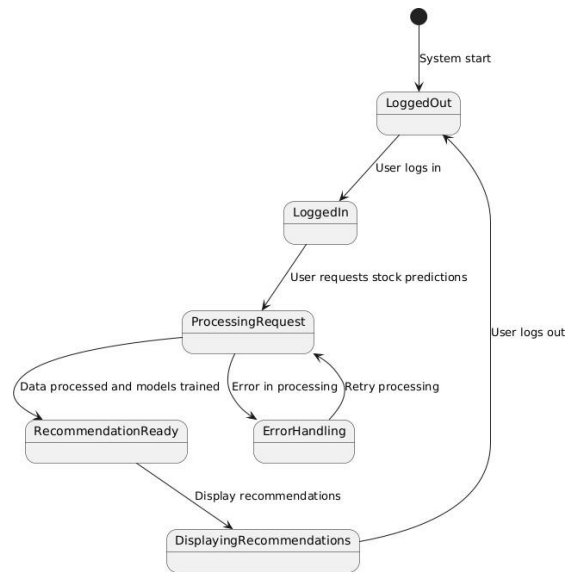


Figure 3.4: Enter Caption

3.3 Data Design

In the HoshyarInvestAI project, the information domain has information related to stock data, financial reports, market trends, user profiles, and investment strategies. Here's how the system transforms this domain into data structures and how the major entities are stored, processed, and organized as follows:

1. Data Sources

Stock Data: fetches real-time as well as past stock information, including share price, ticker symbols, and company-specific financial data. This information can be done either by fetching the APIs like PSX or through downloaded reports of the company's financial statements. **Market Trends:** All economic trends, world events, and other macroeconomic indicators find their reflection in the security price. **Financial Reports:** PDFs, news articles, and journal sources that carry critical company financials, quarterly reports, and investor sentiment.

2. Data Structures and Entities Major Data Entities:

Investor: It keeps personal data including name, e-mail, investment interest, and risk capacity. Structures Organized in structured data and stored in the database called User Data. Methods include setting investment strategies and viewing portfolios.

Portfolio:

It features live investments, historical performance, as well as information about investment allocation. There are a variety of stocks in every portfolio and every portfolio has an investor behind it. It has been related that it is organized relational data in the User Data

database.

Stock:

Stores attributes concerning the stock name, its Ticker Symbol, and current and historical prices. Each share consists of a portfolio and is susceptible to market movements. This is stored in the Stock Data database, which is optimized for fast lookups and updates.

MarketTrend:

Contains info about trends - bullish, bearish, economic indicators, historical events about the market. It is employed in the extraction of stock price forecasts. They are stored in the Market Trends database, making it possible to retrospectively perform historical trend analysis during model training.

InvestmentStrategy:

Contains attributes including type of strategy-growth, value, level of risk and target returns. Backtested and optimized strategies for the investor. Linked to historical data that has been utilized for simulation and testing Stored within the database of User Data.

FeatureEngineering and AnalysisModel (TFT, RL, EGARCH):

Feature engineering performs the transformation of raw data: encoding and scaling. The transformed data are used for models' training with predictions in alterations of the stock movement and measurements of the risk. All are processed in-memory at runtime, but backed up with historical data and model configurations stored in Financial Reports and Stock Data databases.

3-Data Processing and Storage

Data Collection:

Real-time or periodic calls to the actual stock data or financial reports from external sources such as APIs or document feeds. These are stored in a structured format in a database like Stock Data or Financial Reports.

Data Preprocessing:

The actual analysis mainly begins with preprocessed raw data. In most cases, this preprocessing includes cleaning, normalization, and feature extraction through the FeatureEngineering module. Preprocessed data is resident in-memory only but both backed by the Stock Data and Financial Reports databases.

Model Training and Predictions:

Temporal Fusion Transformer (TFT), Reinforcement Learning (RL), and EGARCH models are trained on the historical stock prices and trends, stored in the Historical Data and Market Trends databases. Predictions like how the stock prices will move and the amount of risks can be sent live back to the investor when processed data is fed to these models for

them to churn out the results back to the investor for the purposes of making a decision.

Data Storage Items (Databases) 1-Stock Data:

Stores the information of stocks in real time and historically in terms of prices of stocks, ticker symbols, and information of a company. It is structurally designed for efficient retrieval as well as updating. Both feature engineering and model training modules use it.

2-Market Trends:

Stores all the information related to the economic indicators, information about world-wide events, and all its historical behavior with respect to the market. As it is a time-series database, analysis over certain periods is enabled.

3-Financial Reports: It is a mix of structured and unstructured data through sources such as company financial reports, news and journal sources. : PDF reports are stored in the database with the use of document storage, but a relational model is used for parsed financial metrics.

4-User Data:

Stores investor profiles, portfolios, and investment strategies. A relational database designed with the goal of secure, personalized access and rapid retrieval.

4-Data Organization and Processing:

Normalization:

Data is normalized in such a way that there are no redundancies and consistency in the entire database. The key relationships that are established include a relation between investors to their portfolios, portfolios to stocks, and stocks to market trends.

Indexing:

Stock prices, market trends, financial reports are indexed by attributes which are used to query fast during model training and real-time analysis.

Scaling and Encoding:

Data is scaled in case of numerical values and encoded in case of categorical variables before the actual model training as part of feature engineering.

Backtesting and Simulations:

Using historical stock data and market trends, there is simulation and validation of the investment strategy whereby it gives an insight to investors about how their strategies would have performed in the past.

Conclusion:

In HoshyarInvestAI, multiple finance sources are fed into structured data entities such as stocks, portfolios, and market trends, and those databases store them. The system prepro-

cesses the data it has, feeds it into TFT, RL, EGARCH machine learning models, and provides the investor with real-time predictions and recommendations. The use of efficient storage and retrieval proves that this system is supposed to make fast and personalized, data-driven decisions for investors.

JSON SCHEMA

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{
  "type": "object",
  "properties": {
    "Investor": {
      "type": "object",
      "properties": {
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        "Name": { "type": "string" },
        "Email": { "type": "string" },
        "Password": { "type": "string" },
        "ExperienceLevel": { "type": "string" },
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      "properties": {
        "AdminID": { "type": "integer" },
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        "Email": { "type": "string" },
        "Password": { "type": "string" }
      }
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      }
    },
    "Recommendation": {
      "type": "object",
      "properties": {
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"RecommendationID": { "type": "integer" },  
"Date": { "type": "string", "format": "date" },  
"RecommendationDetails": { "type": "string" }  
}  
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